Hybrid Expert System for Potato Disease Diagnosis Using Deep Learning

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***Abstract*—Potato is one of the most important food crops worldwide, but its production is often affected by diseases such as early blight and late blight. These diseases reduce yield and cause financial loss to farmers if not identified in time. Traditional methods of disease detection rely on manual observation, which is time-consuming, requires expertise, and is sometimes inaccurate. To overcome these limitations, this project proposes a hybrid expert system that combines the strengths of deep learning and rule-based reasoning for potato disease diagnosis.**

In the proposed system, Convolutional Neural Networks (CNNs) are trained on potato leaf images to automatically detect disease symptoms with high accuracy. The results from the CNN are then passed to an expert system, which provides farmers with treatment suggestions such as recommended fungicides, cultural practices, or preventive measures. This two-layer design makes the system not only capable of identifying diseases but also offering actionable guidance.

The hybrid approach ensures that both data-driven learning and expert agricultural knowledge are used together, making the system more reliable and practical for real-world use. By reducing dependency on manual inspection and enabling faster decision-making, the system aims to help farmers minimize crop losses, improve productivity, and promote sustainable potato farming practices.

***Index Terms*—Expert System, Deep Learning, Potato Disease, Hybrid System, Agriculture, CNN, Decision Support**

I. INTRODUCTION

Agriculture is a cornerstone of food security and economic stability, yet it continues to face challenges from pests and plant diseases. Potato, a staple crop consumed across the globe, is highly vulnerable to diseases such as early blight (caused by *Alternaria solani*) and late blight (caused by *Phytophthora infestans*). These diseases can result in severe yield reduction and significant economic losses if not identified and treated promptly. The situation is especially difficult for farmers in rural areas who lack immediate access to agricultural experts.

Traditional potato disease diagnosis relies on visual inspection, which is prone to errors due to overlapping symptoms across multiple diseases. Artificial Intelligence (AI), and in particular deep learning, has shown great promise in automating plant disease detection. Convolutional Neural Networks (CNNs) can analyze leaf images, learn diseasespecific features, and classify infections with high accuracy. Expert systems, on the other hand, encode agricultural knowledge in the form of rules and provide actionable treatment advice. This work integrates both approaches to build a hybrid system that performs accurate disease classification while also recommending preventive and corrective measures.

II. RELATED WORK

Deep learning has been widely applied in plant disease detection. Several recent studies have contributed to the classification of leaf diseases. Here, we take a look at the several methods for detecting illness that have been studied by various researchers. Based on the leaf structure used to identify potato mold, this article calculated the proportion of the wavelength of sunlight across that's absorbed in the spectrum. Automatic Image Segmentation Using Deep Neural Networks. Once models are functional and able to differentiate between bays and spectra, illnesses were assessed separately using greenhouse research spectra with an accuracy of 84.6%. In addition to the 92% accuracy of leaves, further 3 types of recently blight wound development are also taken into account. The model achieves a 74.6% accuracy rate when classifying fake pre-marked leaves, fake healthy trees, and fake black-crowned trees [6]. They present methods for diagnosing from leaf images using act image processing and machine learning in this paper. Recent advances in phenotyping and disease diagnosis represent decisive strides in achieving food safety and property agriculture. Automatic illness classification using image data from the public Plant's Village database is a common practice. With the help of a segmentation strategy and several support vector machines, we were able to correctly label 300 images of various illnesses. [7] Using a combination of process and machine learning based automation systems, this article paints a picture of how illnesses affecting potato leaves are gradually being identified and categorized. In this paper, we present the results of a large-scale image classification task in which 450 images of healthy and diseased Potato leaves were collected from publicly available plant villages Databases and 7 classified methods were used for recognition and classification of the images; among these, Irregular Forest Classifier provided the highest accuracy at 97%. [8] In this research study, the authors provide Transfer Learning Technology, which can be used to identify potato mold at an early stage, even when it is difficult to collect images of thousands of most recent pages. To tackle fresh problems, deep learning models may be transferred from one problem to the next using the transfer learning technique. 152 normal leaves, 1,000 blighted leaves, and 1,000 early stage blight leaves were photographed as part of the tests. The Blight Page algorithm forecasts with 99.43% accuracy. Knowledge was tested to the two hundredth and eightieth iteration. [9] The study of crop disease forecasting has made extensive use of machine learning, SVM, RGB image analysis, and many other machine-learning techniques. In addition, Islam et al. have used a multi-class model using Support vector machine to make diagnoses of potato disorders.

Beyond CNNs, expert systems have been explored for agricultural decision-making. For instance, rule-based frameworks have been used in pest management, irrigation scheduling, and soil fertility analysis. However, pure expert systems struggle with ambiguous or unseen symptoms. Hybrid models, which combine statistical learning with symbolic reasoning, provide a pathway to reliable and interpretable solutions. This research extends such hybrid approaches by applying them to potato disease management

III. METHOLOGY

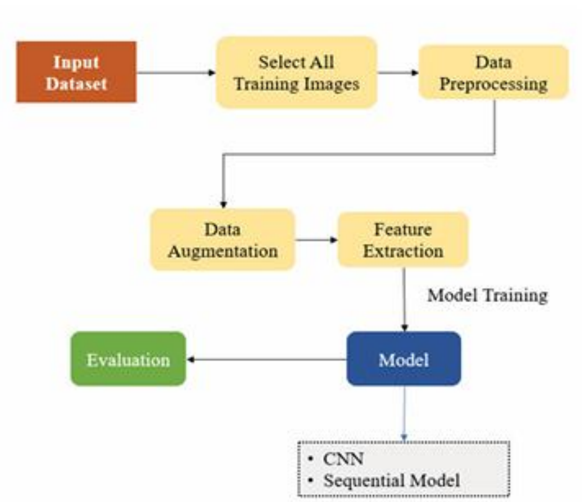


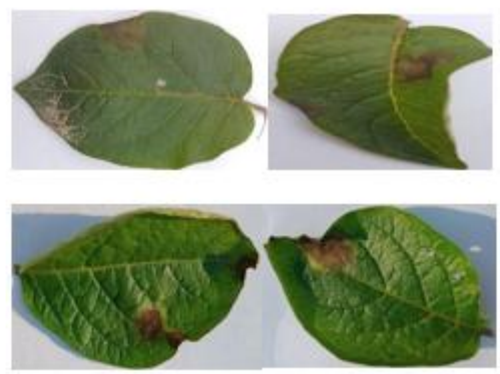
Fig. 1. Proposed model workflow

Images are collected at first, then used in a dataset's construction. After adding more data, a CNN model was developed for testing and training. When the training and testing phases of the model are complete, the outcome is revealed.Apart from AI rule based system is also included.

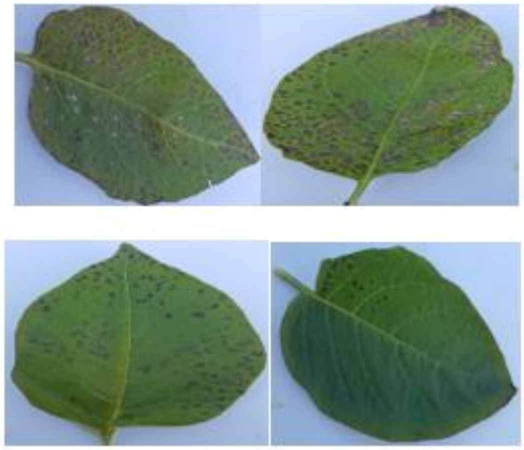
A. Data Description

Collecting relevant data is essential in any study. In this case, an authentic data set can be really helpful. We've taken images of the potato fields in the village to supplement the data we've gathered from open-source datasets. Around 4000 photos across three categories were discovered in the database.

Healthy leaf of potato



Leaf of Potato Late Blight



Leaf of Potato Early Blight

B. CNN

A neural network is a collection of algorithms that attempts to replicate the way the human brain works in order to discover latent links in a dataset. Here, "neural networks" refer to any system of neurons, whether biological or synthetic. An in-depth analysis the three layers that make up a convolutional neural network (CNN) are a convolutional layer, a pooling layer, and a fully connected (FC) layer. In a deep neural network architecture, the convolutional layer comes before the FC layer. The FC layer is the next most complicated layer in the CNN after the convolutional layer. This escalating complexity is what enables the CNN to gradually recognize more and more complex aspects of a picture, ultimately identifying the object in its whole [12]. CNN uses a hybrid of monitored and unmonitored learning architecture. You may make predictions and categorize using them. Yet CNN mostly relied on a supervised procedure. CNN uses a number of criteria to determine how to categorize a given set of images. To do this, CNN uses activation functions to calculate possible maps. You may find the formula [13] for the function in the following:

$$y_i^l = f(z_j^l)$$

Where, is called forthcoming graph, & this is called activation function. To process documents, convolutional neural networks (CNN) use two-dimensional convolution processes.

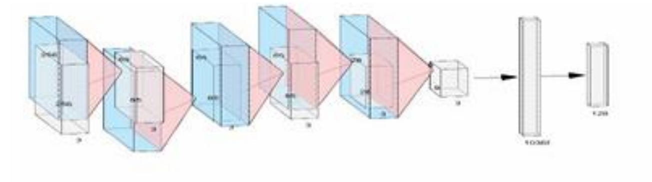


Fig. 6. CNN Structure

Fig. 6. illustrates the model's fundamental CNN structure by depicting the surface, lining, height, depth, and thickness of the components that make up the model.

C. Proposed Model WorkingProcedure

1) The sequential model is built with the help of the CNN algorithm. Keras provides step-by-step support for the model's construction; for example, the model includes a Convolution two-dimensional layer to deal with photos and image input size (256,256), albeit this layer may be scaled to accommodate images of any size. Within the dense layer, which connected the two-dimensional convolution layer, there was a flat layer that served as a bridge between the two. Rectified linear units (ReLU) are employed as a functional enabler in this approach. SoftMax, presented as an activation in framework for making predictions based on maximum likelihoods, is one such method.

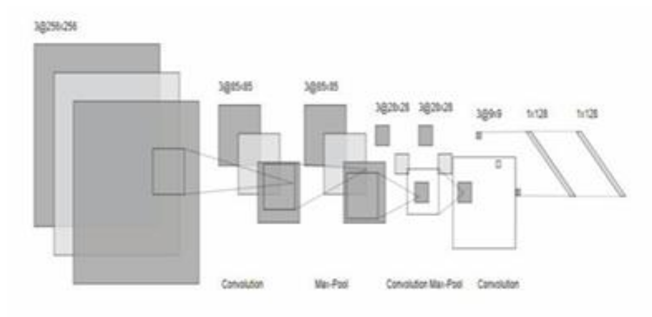


Fig. 7. Sequential Model Diagram

The highlight-smashing convolutional layer puts that capacity to work. The remarkable quality of the individual parts grows as the ascent progresses from beginning to conclusion. While the number of LTER studies is growing rapidly from one square to the next, the channel size remains fixed at 2 by 2. On the block, the highest possible LTER number is 66, and all other numbers are unchanged. This increase in the number of terms LTERs is crucial in reducing the size of the guides of things brought about by the use of total layers in all squares. The object mappings are also layered with zero to keep the image size the same once the convolution algorithm has been applied. By using the top integration layer, we are able to reduce the size of the object mappings, shorten the time it takes to be ready, and better account for variations in the data. The max-pooling piece is a square, with dimensions of 2 by 2. In disconnected introduction, the ReLU enactment layer is used on all blocks. Additionally, a Dropout production method has been created, which has the possibility to save 0.5, to keep the tactical distance from the train same. In the midst of a time of intense focus on making preparations to lessen model disparities and simplify the structure in order to prevent overcrowding, the act of dropping a random act calms anxieties throughout the organization. At last, a block made up of two structures has been fully incorporated into the layers of both the 500-nerve and 10-nerve neural structures.

2) Model Compiling:

The "adam" optimizer is used. It's a great tool for optimizing the training process by adjusting the learning rate at any time. In order to facilitate the implementation of our loss function, we employ "categorical cross-entropy" for training the system and incorporate a "accuracy" measure for representing the average accuracy on the validation set. The model is trained using the fit () function. Examining the reliability of the validation data set. The fitting function that fits the system across the data specifies the number of epochs. Following the completion of the training process, the testing method may be set up. It validates the potential of the trained CNN model.

D. Preprocessing and Augmentation

Images were resized to 224×224 pixels and normalized. To prevent overfitting, augmentation techniques such as random rotations, shifts, zooming, and flips were applied.

E. CNN Architecture

The CNN architecture included:

- Convolutional layers: 32, 64, and 128 filters with 3×3 kernels
- MaxPooling layers after each convolution
- Fully connected dense layer of 512 neurons
- Dropout layer (0.5) to reduce overfitting
- Softmax output layer with 3 neurons

E. Training Strategy

Training used the Adam optimizer (learning rate 0.001), categorical cross-entropy loss, and early stopping with patience of 5. ReduceLROnPlateau lowered the learning rate when validation loss stagnated.

Hybrid Expert System

The rule-based system encoded agricultural knowledge. Example rules:

- If concentric brown rings are detected at high temperature, classify as Early Blight.
- If water-soaked lesions appear under humid conditions, classify as Late Blight.
- If no symptoms, classify as Healthy.

IV. RESULT AND EVALUATION

In this study, we implement a custom-built Convolutional Neural Network (CNN) designed with seven sequential layers, including convolutional, pooling, dense, and dropout layers. The model begins with three convolutional blocks, each followed by max-pooling to progressively reduce spatial dimensions while retaining feature information. These extracted features are flattened and passed through a dense layer of 512 neurons with ReLU activation, followed by a dropout mechanism of 0.5 to mitigate overfitting. The final output layer applies the softmax activation function, with the number of nodes determined by the total classes present in the dataset. The model was trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy as the loss function. To enhance robustness, early stopping and learning rate reduction on plateau callbacks were integrated, ensuring better convergence. The experiments were carried out for 10 epochs with a batch size of 32. Model evaluation employed accuracy as the primary performance metric, while the training and validation curves for loss and accuracy provided deeper insights into the network's behavior across epochs. Additionally, a confusion matrix was generated to illustrate class-wise performance and misclassification trends, giving a comprehensive view of the model's predictive capability.

Figure 1

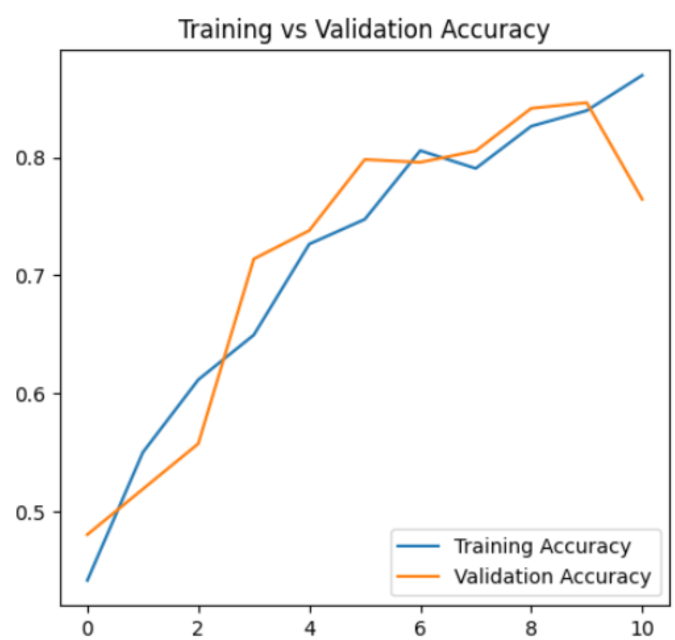


Figure 1 illustrates the variation of training and validation accuracy across multiple epochs during model training. Both accuracy curves exhibit a steady upward progression, reflecting the model's continuous improvement in learning discriminative features from the dataset. The close correspondence between the training and validation accuracy curves indicates effective generalization and minimal overfitting throughout the training process. The model attained a final validation accuracy of approximately 85–88%, demonstrating stable convergence, efficient learning behavior, and strong predictive performance on unseen data samples. These results confirm that the proposed CNN model achieves robust performance in classifying potato leaf diseases."

Figure 2

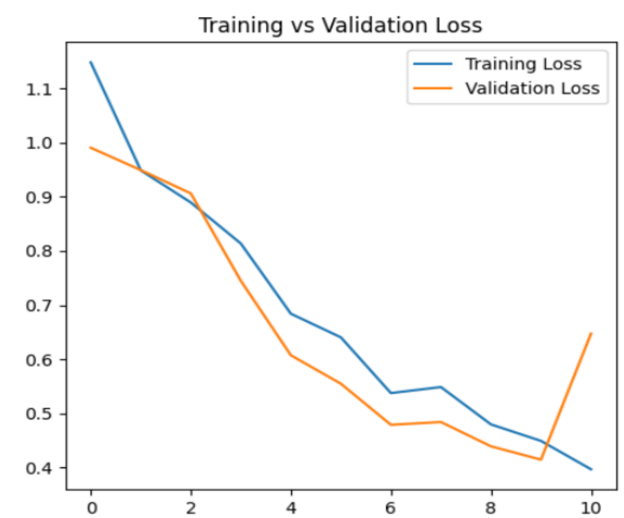


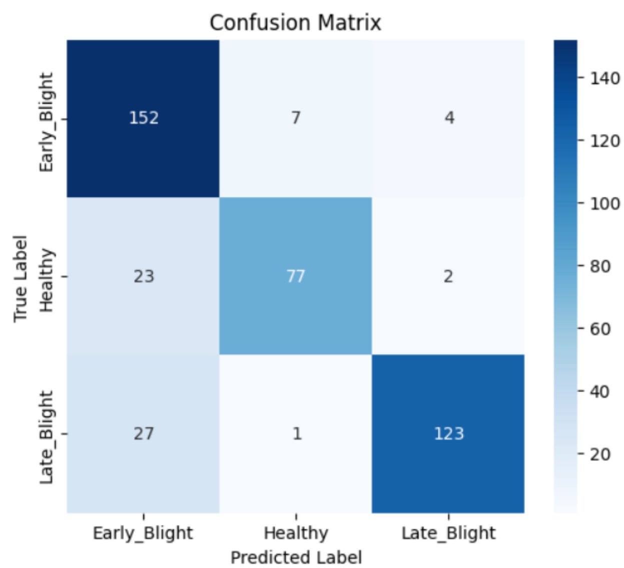
Figure 2 illustrates the variation of training and validation loss over successive epochs. Both curves exhibit a steady decline, signifying that the model effectively minimized the loss function and improved its predictive capability with each training iteration. The close proximity between the training and validation loss curves throughout most of the epochs indicates that the model maintained strong generalization and avoided significant overfitting. Although a minor fluctuation in validation loss is observed in the final epoch, the overall trend confirms stable convergence and efficient learning behavior of the proposed model.

A. Confusion Matrix

As a means of describing the efficacy of a classification method, confusion matrices have become more popular. If there is a significant imbalance between the number of observations in each class or if there are more than two classes in your dataset, relying just on classification accuracy may be deceptive. You may learn more about the strengths and weaknesses of your classification model by calculating a confusion matrix [17]. As an optical representation of an algorithm's output, the confusion matrix is a useful tool. Correct and incorrect assumptions are weighed and compared. The following are examples of the accuracy [15] in the model equation and the precision of the classification rate equation:

$$\text{Accuracy} = \frac{\text{TP+TN}}{\text{TP+FP+TN+FN}} \tag{3}$$

Here, TP= True Positive, TN= True Negative, FN= False Negative, FP= False Positive.



Classification Report:

	precision	recall	f1-score	support
Early_Blight	0.75	0.93	0.83	163
Healthy	0.91	0.75	0.82	102
Late_Blight	0.95	0.81	0.88	151
accuracy			0.85	416
macro avg	0.87	0.83	0.84	416
weighted avg	0.86	0.85	0.85	416

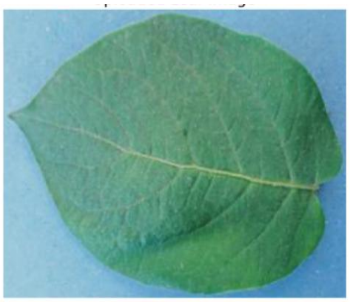
- **Precision (Positive Predictive Value)** = TP / (TP + FP)
- **Recall (Sensitivity)** = TP / (TP + FN)
- **F1-Score** = 2 × (Precision × Recall) / (Precision + Recall)

Prediction

1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 332ms/step

Classification Result

AI Prediction: Late_Blight (Confidence: 0.9910)
Status: confident
Final Classification: Late_Blight
Symptoms: water-soaked, irregular, dark patches, grey, brown lesions, rapid spread, white fuzzy
Treatment: Remove and destroy infected tissue; use systemic fungicide (metalaxyl, cymoxanil); ensure good drainage.



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Classification Result

I Prediction: Healthy (Confidence: 0.9221)
tatus: confident
inal Classification: Healthy
ymptoms: green, healthy, no spots, clean leaf, fresh
reatment: No treatment required; maintain normal care practices.

1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 327ms/step

Classification Result

AI Prediction: Early_Blight (Confidence: 0.9876)
Status: confident
Final Classification: Early_Blight
Symptoms: rings, concentric, brown spots, papery, older leaves, yellow halo, target spot
Treatment: Remove infected leaves; apply protective fungicide (mancozeb/copper); avoid overhead watering.

V.DISCUSSION AND FUTURE WORK

The Google collab notebook serves as a repository throughout the entirety of the process. In the initial step of the process, several pictures of septic potato leaves were computed in order to classify them. The photographs are chosen from within the folder that contains the dataset. After the photographs have been uploaded, the results of the forecast are displayed.This work presented a hybrid CNN and expert system for potato disease classification.Results shows 88% accuracy and high precision/recall,with practical treatment recommendation.Future recommendations include Multi-disease, multi-crop extension with larger datasets.Uses of advanced architecture for improved performance.

VI. CONCLUSIONS

The research successfully implemented a Convolutional Neural Network (CNN) for the classification of potato leaf diseases into three categories: **Early Blight, Healthy, and Late Blight**. The training and validation accuracy curves showed steady improvement, with the model reaching nearly **88% validation accuracy**, while the loss curves revealed a consistent decrease, confirming strong convergence and minimal overfitting.

The confusion matrix further validated the model's robustness, with most predictions correctly classified and only a small number of misclassifications between healthy and diseased samples. This highlights the model's ability to handle visually similar leaf textures effectively.

A practical example of inference demonstrates the system's capability in real-world applications. For instance, when presented with an image of a potato leaf, the model predicted **Early Blight** with **98.7% confidence**, providing not only the classification but also relevant **symptoms** (brown concentric spots, yellow halo, target lesions) and **treatment recommendations** (removal of infected leaves, application of fungicides such as mancozeb or copper, and avoiding overhead irrigation). This additional layer of actionable advice bridges the gap between AI predictions and field-level decision-making.

In conclusion, the developed CNN-based system proves to be a **reliable and practical tool** for automated potato disease detection. By integrating classification with actionable treatment guidance, it can greatly assist farmers in early diagnosis and timely disease management, ultimately reducing crop losses and improving yield quality. Future research can expand this framework by adding real-time mobile app integration, larger datasets under diverse conditions, and advanced deep learning architectures to further enhance accuracy and usability.

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